MODULE 1 TISSUES

Histology: The Four Tissue Types.

TISSUES & HISTOLOGY

Tissues are groups of similar cells performing a specific function. Four types build every organ in the body, and the arrangement of the cells is what predicts what the tissue can do.

COMPETENCIES W1-GERM-LAYERS, W1-EPITHELIAL-ID, W1-CELL-JUNCTIONS, W1-CONNECTIVE-ID, W1-BODY-MEMBRANES

BY THE END

1. Name the four tissue types, their general functions, and where each is found.
2. Define a biopsy and walk through the procedure.
3. Identify the five cell junctions, their proteins, locations, and functions.
4. Separate epithelial from connective tissue on structure alone.
5. Classify epithelium by layers and cell shape, and name a location for each type.
6. Classify glands structurally and functionally.
7. Name the cells, fibers, and ground substance of connective tissue and what each contributes.
8. Recognize cartilage, bone, muscle, and nervous tissue at survey level.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Tissues

drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html

The Four Major Tissue Types

1. **Epithelial tissue**, covers body surfaces, lines cavities, and forms glands.
2. **Connective tissue**, protects, supports, and binds structures and organs.
3. **Nervous tissue**, detects changes and generates nerve impulses.
4. **Muscle tissue**, facilitates movement and generates force.

Cells that stay and cells that move

Most cells are **stationary**. Some **migrate**, particularly during growth and development. White blood cells migrating to a site of injury or infection are the everyday example.

Biopsy

Definition. Removal of a tissue sample for microscopic examination. This is how tissue-level questions get answered clinically.

1. Collect a tissue sample using a needle, scalpel, or endoscope.
2. Preserve the tissue in a fixative such as formalin.
3. Section the tissue into thin slices for examination.
4. Stain the tissue for contrast under a microscope.

The Five Cell Junctions

JUNCTION, PROTEIN, LOCATION, FUNCTION

Junction	What it is	Function	Location	Major protein
Tight junction	A network of transmembrane proteins that fuse together on adjacent plasma membranes, sealing the passageway between cells	Prevents substances passing between cells, prevents organ contents leaking into surrounding tissue	Epithelium lining the stomach, intestines, urinary bladder, blood-brain barrier	Claudins and occludins
Adherens junction	Adhesion belts formed by transmembrane proteins attached to microfilaments	Provides mechanical stability, helps epithelial surfaces resist separation during contractile activity	Intestinal epithelial cells	Cadherins
Desmosome	Plaques and cadherins connected to intermediate filaments	Strongly binds cells to each other, prevents separation under tension	Outer epidermis, cardiac muscle cells	Cadherins
Hemidesmosome	Anchors cells to the basement membrane	Prevents separation of epidermis from basement membrane, stabilizes epithelial cells during mechanical stress	Skin, heart muscle	Integrins
Gap junction	Protein channels connecting neighboring cells, allowing passage of small ions and molecules	Cell to cell communication, transfers nutrients and waste, allows impulses to spread	Lens, cornea, embryonic tissues, neurons, cardiac tissue, GI tract, uterus	Connexins

HOLD ONTO THIS

Desmosomes connect cell to cell. Hemidesmosomes connect cell to basement membrane. Half the name, half the job.

Epithelium versus Connective Tissue

THE TWO TISSUE TYPES COMPARED

Feature	Epithelial tissue	Connective tissue
Matrix and cells	Tightly packed cells, minimal extracellular matrix	Large extracellular matrix, few cells
Blood vessels	Avascular, relies on diffusion	Highly vascular, with cartilage and tendon as exceptions
Location	Covers surfaces and forms linings	Supports and connects other tissues

Similarities. They work together for nutrient exchange, since avascular epithelium depends on the vascular connective tissue beneath it. Both play critical roles in maintaining structure and function.

Epithelial Tissue

Three functions of epithelium

- **Selective barrier**, controls transfer of substances into and out of the body.
- **Secretory surface**, produces and releases cellular products.
- **Protective surface**, provides resistance against abrasion and environmental influences.

The basement membrane

Location. Below the basal surface of epithelial tissue. **Composition.** A **basal lamina** of collagen, laminin, and glycoproteins, plus a **reticular lamina** of collagen and glycoproteins.

Functions. Supports and anchors epithelial tissue, facilitates wound healing and cell migration, restricts large molecules, and filters substances in the kidneys.

Classification of epithelium

EPITHELIAL TYPES, FUNCTION, AND LOCATION

Tissue type	Description	Function	Locations
Simple squamous	Single layer of flat, thin cells	Filtration, diffusion, osmosis, secretion	Alveoli, glomeruli, lining of blood vessels, serosa
Simple cuboidal	Single layer of cube-shaped cells	Secretion and absorption	Kidney tubules, small glands, ovary surface
Simple columnar	Single layer of tall, column-shaped cells, may have microvilli or goblet cells	Absorption, secretion of mucus and enzymes	Digestive tract from stomach to rectum, gallbladder
Stratified squamous	Several layers, apical layers are flat	Protection against abrasion	Skin (keratinized), esophagus, mouth, vagina
Stratified cuboidal	Two or more layers of cube-shaped cells	Protection and limited secretion	Sweat glands, mammary glands, salivary glands
Stratified columnar	Several layers, apical cells are columnar	Protection and secretion	Male urethra, ducts of some glands
Transitional	Several layers, cells stretch and change shape from dome-shaped to flat	Stretching to allow distension	Urinary bladder, ureters, part of the urethra
Pseudostratified ciliated columnar	Appears stratified but all cells touch the basement membrane, may have cilia and goblet cells	Secretion of mucus, propulsion by ciliary action	Respiratory tract (trachea, bronchi), sperm ducts

HOLD ONTO THIS

Name the layers first, then the shape of the apical cells. Simple means one layer, stratified means more than one, and the shape you name is always the shape at the free surface.

Glandular Epithelium

Structural classification

GLANDS BY NUMBER OF CELLS

Type	Description	Example
Unicellular	Single-celled glands	Goblet cells, which secrete mucus
Multicellular	Multiple cells forming a structure	Salivary glands, sweat glands

Duct type. Simple means no branching. Compound means branching. **Shape of the secretory portion.** Tubular means tube-shaped. Acinar means rounded. Combine the two and you have the full structural name.

Endocrine and exocrine

Endocrine glands release their product into the bloodstream, with no duct. Thyroid, adrenal, pituitary. **Exocrine glands** secrete their products into ducts that empty onto a surface or into a lumen. Sweat, salivary, sebaceous. Some glands are **mixed**, doing both, and the pancreas is the classic example: exocrine digestive enzymes into a duct, endocrine insulin and glucagon into the blood.

Functional classification of exocrine glands

HOW THE PRODUCT ACTUALLY LEAVES THE CELL

Type	Mechanism	Example
Merocrine	Secretions released via exocytosis. The cell is unharmed. Most common type	Salivary glands, pancreas
Apocrine	Secretions accumulate at the apical surface, which pinches off to release the product. The cell then repairs itself	Mammary glands
Holocrine	Secretions accumulate in the cytoplasm and the entire cell ruptures to release the product	Sebaceous glands

MEMORY TOOL

HOL equals WHOLE cell is secreted.

Connective Tissue

Functions

- Binds together, supports, and strengthens body tissues.
- Protects and insulates internal organs.

- Compartmentalizes structures, as in skeletal muscle.
- Serves as a major transport system. Blood is a connective tissue.
- Is the primary location of stored energy reserves, as in adipose tissue.
- Is a main source of immune responses.

The two basic elements

Connective tissue is **extracellular matrix** plus **cells**. The matrix divides into **ground substance** and **protein fibers**. The proportions of these elements are what make blood liquid, cartilage firm, and bone hard.

Ground substance

Definition. The material between cells and protein fibers. **Structure.** Fluid, semifluid, gelatinous, or calcified. **Composition.** Water and organic molecules such as chondroitin sulfate and glucosamine.

Functions. Supports and binds cells, supplies water and nutrients for exchange between blood and cells, and plays a role in tissue development and metabolism.

The matrix determines the quality. Cartilage is firm but pliable. Bone is hard and inflexible. Same basic plan, different matrix.

The three fiber types

COLLAGEN, RETICULAR, AND ELASTIC FIBERS

Fiber	Protein	Properties	Locations and functions
Collagen	Collagen, the most abundant protein in the body at about 25 percent	Very strong, resists tension. Parallel bundles provide tensile strength	Most connective tissues: bone, tendon, cartilage
Reticular	Collagen covered with glycoproteins	Fine, branching networks	Walls of blood vessels, surrounding cells in areolar, adipose, nervous, and smooth muscle tissue. Provides support and strength, forms the stroma of soft organs, helps form the basement membrane
Elastic	Elastin, with the glycoprotein fibrillin providing tensile strength and stability	Stretches up to 150 percent of relaxed length and returns to original shape	Skin, lungs, blood vessel walls

Cells of connective tissue

WHO LIVES IN THE MATRIX

Cell	Origin or definition	Function and location
Fibroblasts	Found in all general connective tissues	Migrate through connective tissue secreting matrix fibers and ground substance
Macrophages	Irregular white blood cells derived from monocytes	Engulf bacteria and debris via phagocytosis. Fixed macrophages stay in one tissue such as liver, lungs, spleen. Wandering macrophages migrate to sites of infection or inflammation
Mast cells	Found along blood vessels supplying connective tissue	Secrete histamine, which dilates blood vessels during inflammation. Also bind, ingest, and kill bacteria
Plasma cells	Develop from B lymphocytes	Secrete antibodies. Found in connective tissue of the GI tract, respiratory tract, and lymphatic tissues
Adipocytes	Connective tissue cells that store triglycerides	Found deep in the skin and around organs such as the heart and kidneys
White blood cells	Various leukocytes	Immune defense

Blasts versus cytes

IMMATURE AND MATURE FORMS

	Immature, the blasts	Mature, the cytes
Examples	Fibroblasts, osteoblasts, chondroblasts	Fibrocytes, osteocytes, chondrocytes
What they do	Actively divide and secrete extracellular matrix	Maintain and repair extracellular matrix

General features of connective tissue

- **Location.** Found below epithelial tissue, never on body surfaces.
- **Vascularity.** Highly vascular. Exceptions: cartilage and tendon.
- **Innervation.** Supplied with nerves. Exceptions: cartilage and blood.

Embryonic mesenchyme and mucous connective tissue

THE TWO EMBRYONIC CONNECTIVE TISSUES

	Embryonic mesenchyme	Mucous connective tissue
Location	In the embryo, under skin and along bones	In the umbilical cord
Significance	Differentiates into all types of connective tissue	Provides support
Characteristics	Irregular cells in a semifluid matrix with delicate reticular fibers	Widely scattered fibroblasts in a jelly-like matrix

Blood, a Fluid Connective Tissue

FORMED ELEMENTS AND WHAT EACH DOES

Cell type	Function
Neutrophils	Phagocytosis of bacteria
Eosinophils	Combat parasitic infections and allergies
Lymphocytes	Fight viruses
Monocytes and macrophages	Phagocytosis, transform into macrophages
Basophils	Involved in allergic responses
Red blood cells	Transport oxygen and carbon dioxide
Platelets	Facilitate blood clotting

Connective Tissues, Cartilage, Bone, and Nervous Tissue

SURVEY OF THE REMAINING TISSUE TYPES

Tissue type	Description	Function	Locations
Areolar connective	Loose arrangement of collagen, elastic, and reticular fibers with scattered cells	Strength, elasticity, support	Under epithelial tissue, around organs and blood vessels
Adipose connective	Cells filled with fat droplets, sparse extracellular matrix	Energy storage, insulation, cushioning	Subcutaneous tissue, around kidneys and heart
Hyaline cartilage	Glassy appearance, fine collagen fibers not visible with ordinary staining, chondrocytes in lacunae	Smooth surface for joint movement, flexible support	Ends of long bones, costal cartilage, nose, trachea, larynx
Fibrocartilage	Thick collagen fibers dominate	Resists compression, absorbs shock	Intervertebral discs, pubic symphysis, knee menisci
Elastic cartilage	Similar to hyaline but with more elastic fibers	Maintains shape and provides flexibility	External ear (pinna), epiglottis
Compact bone	Hard, calcified matrix with collagen fibers, osteocytes in lacunae	Supports, protects, stores calcium, houses blood formation	Bones
Nervous tissue	Neurons and glial cells, branching neurons with long processes	Transmission of electrical signals	Brain, spinal cord, nerves

Muscle Tissue

THE THREE MUSCLE TISSUE TYPES

Tissue type	Description	Function	Locations
Skeletal muscle	Long, cylindrical, multinucleated cells, striated	Voluntary movement, heat production	Attached to bones and occasionally skin
Cardiac muscle	Branching cells connected by intercalated discs, striated, uninucleated	Involuntary contraction to pump blood	Heart walls
Smooth muscle	Spindle-shaped cells, non-striated, uninucleated	Involuntary movement, propels substances	Walls of hollow organs such as intestines and bladder

TISSUE STRUCTURE PREDICTS HOW IT HEALS AND HOW IT FAILS

Vascularity sets healing speed. Cartilage and tendon are avascular or nearly so, which is why a torn meniscus or a ruptured Achilles heals slowly while a skin wound closes in days. Junction type sets what tears. Loss of desmosomal adhesion in the epidermis produces blistering, because the cells separate from each other. Loss of hemidesmosomes separates the whole epidermis from the basement membrane, which is a deeper and more serious blister. Cell arrangement sets the barrier. Stratified squamous epithelium survives abrasion because it sheds its top layer. Simple squamous cannot, which is why the alveoli, one cell thick for gas exchange, are so easily damaged.